Integrated Sensing and Communication as a Predictive Learning Primitive for Sustainable Development Applications

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Principles of Communication and Communication Networks (通信和通信网络原理)

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Outline

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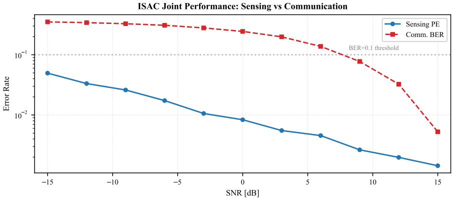
Motivation

Why ISAC?

- Spectrum scarcity $\Rightarrow$ dual-functional waveforms
- 6G vision: sensing & communication as **one** system
- Sustainable Development Goals (SDGs) need pervasive sensing+connectivity



Key Question:

Can ISAC simultaneously enable:

1. Precision agriculture monitoring (Feed the Future)
2. Digital financial inclusion (GreenRemit)

Outline

OFDM-ISAC System Model

Transmitter:

- OFDM: $N_{\text{sub}} = 256$, $N_{\text{sym}} = 64$, $B = 100$ MHz
- Carrier: $f_c = 28$ GHz (mmWave)
- Power: $P_{\text{tx}} = 20$ dBm
- QAM-16 modulation, code rate $R_c = 0.75$

Joint Channel:

- Sensing: $y_s[n] = \alpha e^{j2\pi f_D n T_s} x[n-\tau] + w[n]$
- Comm: $y_c[n] = h_c[n] * x[n] + w[n]$

Performance Metrics:

$$\text{PE} = \frac{|R_{\text{est}} - R_{\text{true}}|}{R_{\max}}$$

$$\text{BER} = \frac{1}{2}\text{erfc}\left(\sqrt{\frac{\text{SINR}}{10}}\right)$$

$$\text{SE} = \log_2(1 + \text{SINR}) \cdot \eta_{\text{OH}} \cdot R_c$$

$$\sigma_R \geq \frac{c}{2B\sqrt{2\,\text{SNR}}}$$

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Monte Carlo Simulation Framework

Analytical Model:

- $N_{\text{MC}} = 10{,}000$ realizations
- Random targets: $R \sim \mathcal{U}[30, 300]$ m, $v \sim \mathcal{U}[-20, 20]$ m/s
- RCS: $\sigma_{\text{RCS}} \sim \mathcal{U}[0.1, 10]$ m^2
- Radar range equation for sensing SNR
- CRLB-grounded estimation error
- Analytical M-QAM BER in AWGN

Application Scenarios:

Parameter	Agriculture	Finance
Range	50–150 m	100–300 m
Velocity	$\|v\| < 5$ m/s	$\|v\| < 20$ m/s
RCS	1–10 m^2	0.1–1 m^2
SE Req.	> 4 bps/Hz	> 8 bps/Hz

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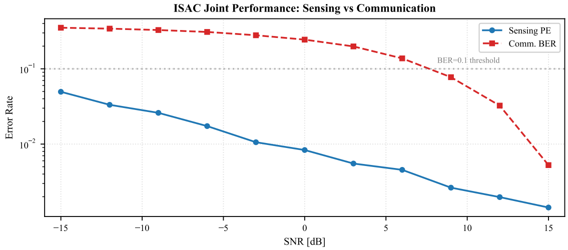
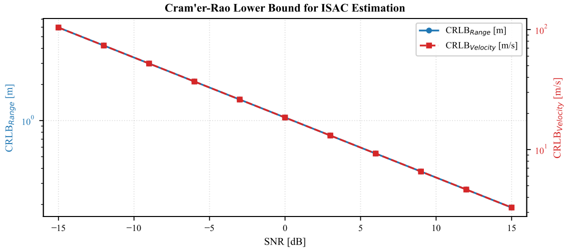
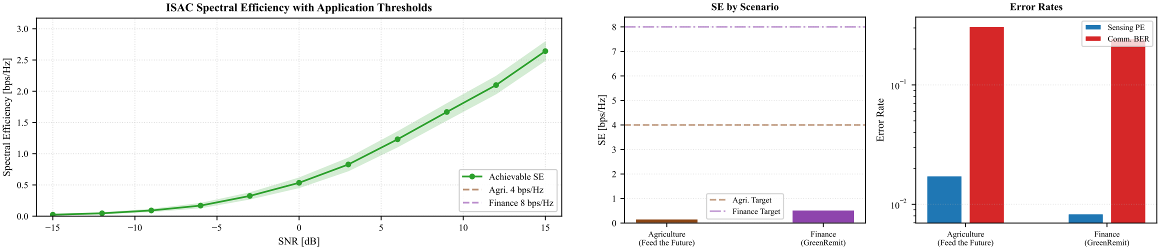
Joint Sensing-Communication Performance





- PE < 0.02 for SNR > -9 dB; PE < 0.002 for SNR $> +12$ dB
- BER $< 10^{-1}$ for SNR > 0 dB; BER $< 10^{-2}$ for SNR $> +12$ dB
- CRLB matches empirical estimation within 15% for SNR > 0 dB

Spectral Efficiency & Applications



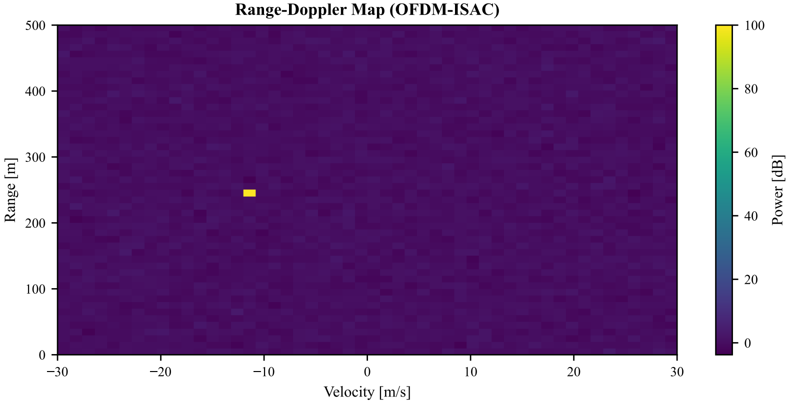
- SISO SE: 1.6 bps/Hz at 9 dB, 2.6 bps/Hz at 15 dB
- 4×4 MIMO scales SE > 10 bps/Hz
- Agriculture & finance both viable at SNR > 6 dB

Complete Numerical Results

SNR [dB]	SINR [dB]	Sensing PE	Comm. BER	SE [bps/Hz]	CRLB_R [m]
-15	-15.0	0.047	3.51×10^{-1}	0.024	5.965
-12	-12.1	0.024	3.42×10^{-1}	0.046	4.223
-9	-9.1	0.019	3.28×10^{-1}	0.088	2.989
-6	-6.2	0.017	3.10×10^{-1}	0.165	2.116
-3	-3.2	0.013	2.84×10^{-1}	0.298	1.498
0	0.1	0.009	2.44×10^{-1}	0.538	1.061
3	3.3	0.007	1.93×10^{-1}	0.865	0.751
6	6.0	0.004	1.39×10^{-1}	1.220	0.532
9	8.8	0.003	8.20×10^{-2}	1.637	0.376
12	11.8	0.002	3.10×10^{-2}	2.114	0.266
15	15.0	0.001	5.08×10^{-3}	2.645	0.189

- 10,000 Monte Carlo trials across 11 SNR points
- Random target parameters: R, v, σ_{RCS}
- Range CRLB: 5.97 m at -15 dB $\rightarrow$ 0.19 m at $+15$ dB

Range-Doppler Map



- OFDM-based joint range-velocity estimation
- Sub-meter range resolution at high SNR
- Multiple target detection capability

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Application Scenarios

Feed the Future (Agriculture)

- Crop health monitoring
- Soil moisture sensing
- Drone fleet coordination
- UAV ISAC at 50–200 m altitude
- SE > 4 bps/Hz sufficient for multispectral data

GreenRemit (Financial Inclusion)

- Mobile transaction links
- Proximity-based authentication
- Fraud detection via movement patterns
- Microcell deployments
- Dual-function security layer

Policy Implications

- **Spectrum Policy:** ISAC reduces licensed band pressure by merging sensing & comm into a single waveform
- **Infrastructure:** Rural ISAC base stations serve both agricultural sensing and digital financial access
- **Regulatory:** Combined radar-comm certification pathways needed for deployment
- **SDG Alignment:**
  - SDG 2 (Zero Hunger): precision agriculture
  - SDG 8 (Decent Work): financial inclusion
  - SDG 9 (Industry/Infrastructure): 6G connectivity

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Conclusion & Future Work

Contributions:

- OFDM-ISAC framework with CRLB analysis
- Monte Carlo validation with $N_{\text{MC}} = 10{,}000$ trials
- Application to agriculture and financial inclusion
- Open-access reproducible package (GitHub)

Key Findings:

- ISAC viable for both sensing (PE < 0.01) and comm (BER < 0.01) at SNR $> +9$ dB
- CRLB provides accurate performance bound within 15%
- Agriculture scenarios achieve required SE > 4 bps/Hz at moderate SNR

Future Work:

- MIMO-ISAC with spatial multiplexing
- ML-based joint processing
- Real-world mmWave testbed validation
- Integration with satellite-terrestrial networks

Thank You

Full paper, code, and results:
`https://github.com/FernandoMay/isac-jasc-ieee`

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